Problem Set 69

How Derivatives Affect the Shape of a Graph

April 27, 2026

Problem 2

The graph of the derivative of f' of a function f is shown in Figure 2.

(a) On what intervals is f increasing or decreasing?

$$f'(x) > 0, \quad \forall x \in (1,5)$$
$$f'(x) < 0, \quad \forall x \in (0,1) \cup (5,6)$$

$f(x)$ is increasing on $(1,5)$ because $f'(x) > 0$ on this interval.

$f(x)$ is decreasing on $(0,1)$ and $(5,6)$ because $f'(x) > 0$ on these intervals.

(b) At what values of x does f have a local maximum or minimum?

$$f'(x) = 0 \implies x = 1 \text{ or } 5$$

$f'(x)$ changes from negative to positive at $x = 1$. Therefore, $f(x)$ has a local minimum at $x = 1$.

$f'(x)$ changes from positive to negative at $x = 5$. Therefore, $f(x)$ has a local maximum at $x = 5$.

Problem 4

Find the local maximum and minimum values of the function $g(x) = x + 2\sin x,\ 0 \leq x \leq 2\pi$.

$$g'(x) = 2\cos x + 1$$
$$g'(x) = 0 \implies \cos x = -\frac{1}{2}$$
$$x = \frac{2\pi}{3} \text{ or } \frac{4\pi}{3}$$
$$g'(x) > 0, \quad \forall x \in [0, \frac{2\pi}{3}) \cup (\frac{4\pi}{3}, 2\pi]$$
$$g'(x) < 0, \quad \forall x \in (\frac{2\pi}{3}, \frac{4\pi}{3})$$

$g'(x)$ changes from positive to negative at $x = \frac{2\pi}{3}$. Therefore, $g(x)$ has a local maximum at $x = \frac{2\pi}{3}$.

$g'(x)$ changes from negative to positive at $x = \frac{4\pi}{3}$. Therefore, $g(x)$ has a local minimum at $x = \frac{4\pi}{3}$.

$$g(\frac{2\pi}{3}) = \frac{2\pi}{3} + 2\sin\frac{2\pi}{3} = \frac{2\pi}{3} + \sqrt{3}$$

$$g(\frac{4\pi}{3}) = \frac{4\pi}{3} + 2\sin\frac{4\pi}{3} = \frac{4\pi}{3} - \sqrt{3}$$
$$\boxed{\max : \frac{2\pi}{3} + \sqrt{3}, \min : \frac{4\pi}{3} - \sqrt{3}}$$

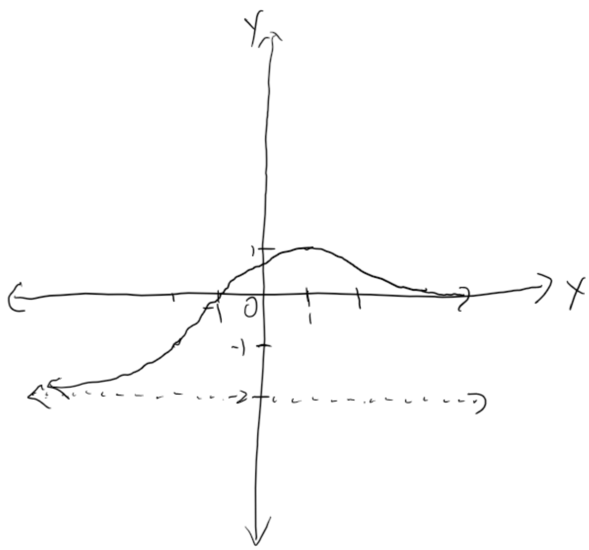
Problem 5

Sketch a possible graph of a function f that satisfies the following conditions:

(i) $f'(x) > 0$ on $(-\infty, 1)$, $f'(x) < 0$ on $(1, \infty)$

(ii) $f''(x) > 0$ on $(-\infty, -2)$ and $(2, \infty)$, $f''(x) < 0$ on $(-2, 2)$

(iii) $\lim_{x\to-\infty} f(x) = -2$, $\lim_{x\to\infty} f(x) = 0$



Problem 6

For each of the functions given below:

- Find the intervals on which f is increasing or decreasing.
- Find the local maximum and minimum values of f.
- Find the intervals of concavity and the inflection points.

(b) $f(x) = e^{2x} + e^{-x}$

$$f'(x) = 2e^{2x} - e^{-x}$$
$$f'(x) = 0 \implies 2e^{2x} = e^{-x}$$
$$\ln 2 + 2x = -x$$
$$x = -\frac{\ln 2}{3}$$

$$f'(x) > 0, \quad \forall x > -\frac{\ln 2}{3}$$
$$f'(x) < 0, \quad \forall x < -\frac{\ln 2}{3}$$

$f(x)$ is increasing on $(-\frac{\ln 2}{3}, \infty)$ because $f'(x) > 0$ on this interval.

$f(x)$ is decreasing on $(-\infty, -\frac{\ln 2}{3})$ because $f'(x) < 0$ on this interval.

$f'(x)$ changes from negative to positive at $x = -\frac{\ln 2}{3}$. Therefore, $f(x)$ has a local minimum at $x = -\frac{\ln 2}{3}$.

$$f(-\frac{\ln 2}{3}) = e^{-\frac{2\ln 2}{3}} + e^{\frac{\ln 2}{3}} = \frac{1}{\sqrt[3]{4}} + \sqrt[3]{2}$$
$$\boxed{\text{min} : \frac{1}{\sqrt[3]{4}} + \sqrt[3]{2}}$$
$$f''(x) = 4e^{2x} + e^{-x}$$
$$f''(x) > 0$$

$f(x)$ is concave up on $\mathbb{R}$ because $f''(x) > 0$ on $\mathbb{R}$.

There are no inflection points.

(c) $f(x) = \dfrac{\ln x}{\sqrt{x}}$

$$x > 0$$
$$f'(x) = \frac{\frac{\sqrt{x}}{x} - \frac{\ln x}{2\sqrt{x}}}{x} = \frac{2 - \ln x}{2x\sqrt{x}}$$
$$f'(x) = 0 \implies \ln x = 2 \implies x = e^2$$
$$f'(x) > 0, \quad \forall 0 < x < e^2$$
$$f'(x) < 0, \quad \forall x > e^2$$

$f(x)$ is increasing on $(0, e^2)$ because $f'(x) > 0$ on this interval.

$f(x)$ is decreasing on (e^2, ∞) because $f'(x) < 0$ on this interval.

$f'(x)$ changes from positive to negative at $x = e^2$. Therefore, $f(x)$ has a local maximum at $x = e^2$.

$$f(e^2) = \frac{\ln e^2}{\sqrt{e^2}} = \frac{2}{e}$$
$$\boxed{\text{max} : \frac{2}{e}}$$
$$f''(x) = \frac{(\ln x - 2)3\sqrt{x} - 2\sqrt{x}}{4x^3}$$

$$f''(x) = 0 \implies 3\ln x = 8 \implies x = e^{\frac{8}{3}}$$
$$f''(x) > 0, \quad \forall x > e^{\frac{8}{3}}$$
$$f''(x) < 0, \quad \forall 0 < x < e^{\frac{8}{3}}$$

$f(x)$ is concave up on $(e^{\frac{8}{3}}, \infty)$ because $f''(x) > 0$ on this interval.

$f(x)$ is concave down on $(0, e^{\frac{8}{3}})$ because $f''(x) < 0$ on this interval.

Inflection point: $\boxed{x = e^{\frac{8}{3}}}$

Problem 7

Find the local maximum and minimum values of f using both the First and Second Derivative Tests. Which methods do you prefer?

(a) $f(x) = x^5 - 5x + 3$

$$f'(x) = 5x^4 - 5$$
$$f'(x) = 0 \implies x^4 = 1 \implies x = 1 \text{ or } -1$$
$$f'(x) > 0, \quad x \in (-\infty, -1) \cup (1, \infty)$$
$$f'(x) < 0, \quad x \in (-1, 1)$$

$f'(x)$ changes from positive to negative at $x = -1$. Therefore, $f(x)$ has a local maximum at $x = -1$.

$f'(x)$ changes from negative to positive at $x = 1$. Therefore, $f(x)$ has a local minimum at $x = 1$.

$$f''(x) = 20x^3$$
$$f''(1) > 0, f''(-1) < 0$$

$f'(1) = 0$ and $f''(1) > 0$. Therefore, $f(x)$ has a local minimum at $x = 1$.

$f'(-1) = 0$ and $f''(-1) < 0$. Therefore, $f(x)$ has a local maximum at $x = -1$.

$$f(1) = 1^5 - 5 \cdot 1 + 3 = -1$$
$$f(-1) = (-1)^5 - 5 \cdot (-1) + 3 = 7$$
$$\boxed{\text{max} : 7, \text{min} : -1}$$

(b) $f(x) = x + \sqrt{1 - x}$

$$x < 1$$
$$f'(x) = 1 - \frac{1}{2\sqrt{1 - x}}$$
$$f'(x) = 0 \implies 1 = 2\sqrt{1 - x}$$
$$x = \frac{3}{4} \text{ or } 1$$

$$f'(x) > 0, \quad x < \frac{3}{4}$$
$$f'(x) < 0, \quad \frac{3}{4} < x < 1$$

$f'(x)$ changes from positive to negative at $x = \frac{3}{4}$. Therefore, $f(x)$ has a local maximum at $x = \frac{3}{4}$.

$$f''(x) = -\frac{1}{4(1-x)^{\frac{3}{2}}}$$
$$f''(x) < 0$$

$f'(\frac{3}{4}) = 0$ and $f''(\frac{3}{4}) < 0$. Therefore, $f(x)$ has a local maximum at $x = \frac{3}{4}$.

$$f(\frac{3}{4}) = \frac{3}{4} + \sqrt{1 - \frac{3}{4}} = \frac{5}{4}$$
$$\boxed{\text{max} : \frac{5}{4}}$$

Problem 8

The graph of the derivative f' of a continuous function f is shown in Figure 4.

(a) On what intervals is f increasing or decreasing?

$$f'(x) > 0, \quad \forall x \in (3,6) \cup (6,9)$$
$$f'(x) < 0, \quad \forall x \in (0,3)$$

$f(x)$ is increasing on $(3,6) \cup (6,9)$ because $f'(x) > 0$ on this interval.

$f(x)$ is decreasing on $(0,3)$ because $f'(x) < 0$ on this interval.

(b) At what values of x does f have a local maximum or minimum?

$f'(x)$ changes from negative to positive at $x = 4$ and $x = 8$. Therefore, $f(x)$ has local minimums at $x = 4$ and $x = 8$.

$f'(x)$ changes from positive to negative at $x = 2$. Therefore, $f(x)$ has a local maximum at $x = 2$.

(c) On what intervals is f concave upward or downward?

$$f''(x) > 0, \quad \forall x \in (3,6) \cup (6,9)$$
$$f''(x) < 0, \quad \forall x \in (0,3)$$

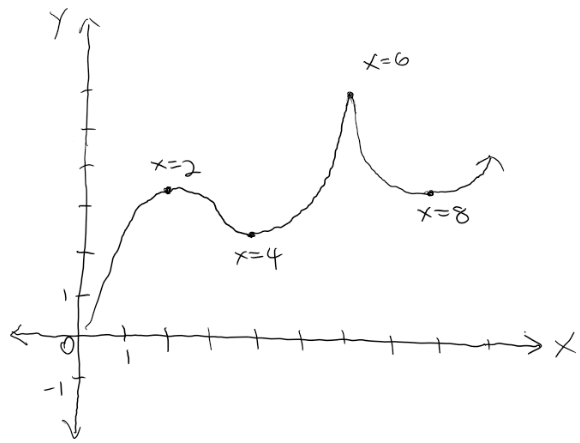
$f(x)$ is concave upward on $(3,6)$ and $(6,9)$ because $f''(x) > 0$ on these intervals.

$f(x)$ is concave downward on $(0,3)$ because $f''(x) < 0$ on this interval.

(d) State the x-coordinate(s) of the point(s) of inflection.

$f(x)$ changes from concave downward to concave upward at $x = 3$. Therefore, $x = 3$ is an inflection point.

(e)



Problem 9

Suppose f'' is continuous on $(-\infty, \infty)$.

(a) If $f'(2) = 0$ and $f''(2) = -5$, what can you say about f?

$f(x)$ has a local maximum at $x = 2$.

(b) If $f'(6) = 0$ and $f''(6) = 0$, what can you say about f?

$f(x)$ has a local maximum, local minimum, or neither at $x = 6$.

Problem 11

Suppose the derivative of a function f is $f'(x) = (x+1)^2(x-3)^5(x-6)^4$. On what interval is f increasing?

$$f'(x) = (x+1)^2(x-3)^5(x-6)^4$$
$$f'(x) = 0 \implies x = -1, 3, 6$$
$$f'(x) > 0, \quad \forall x \in (3,6) \cup (6, \infty)$$
$$f'(x) < 0, \forall x \in (-\infty, -1) \cup (-1, 3)$$

$f(x)$ is increasing on $(3,6)$ and $(6,\infty)$ as $f'(x) > 0$ on these intervals.

Problem 12

Let $K(t)$ be a measure of the knowledge you gain by studying for a test for t hours. Which do you think is larger, $K(8) - K(7)$ or $K(3) - K(2)$? Is the graph of K concave upward or concave downward? Why?

Given the knowledge gained per hour decreases with time. By inspection, we define $K(t)$ as $\ln x$.

$$K'(t) = \frac{1}{x}$$

$$K''(t) = -\frac{2}{x^2}$$
$$\boxed{K(3) - K(2) > K(8) - K(7)}$$

$K(t)$ is concave downward because $K''(t) < 0$.

Problem 13

Find a cubic function $f(x) = ax^3 + bx^2 + cx + d$ that has a local maximum value of 3 at $x = -2$ and a local minimum value of 0 at $x = 1$.

$$f'(x) = 3ax^2 + 2bx + c$$

$$f(-2) = -8a + 4b - 2c + d = 3 \tag{1}$$
$$f(1) = a + b + c + d = 0 \tag{2}$$
$$f'(-2) = 12a - 4b + c = 0 \tag{3}$$
$$f'(1) = 3a + 2b + c = 0 \tag{4}$$

$$(2) - (1) \implies 3a - b + c = -1$$
$$(4) - (2) + (1) \implies 3b = 1 \implies b = \frac{1}{3}$$
$$(4) - (3) \implies 2b = 3a \implies a = \frac{2}{9}$$
$$c = -\frac{4}{3}, d = \frac{7}{9}$$
$$\boxed{f(x) = \frac{2}{9}x^3 + \frac{1}{3}x^2 - \frac{4}{3}x + \frac{7}{9}}$$

Problem 14

Show that the curve $y = \dfrac{1+x}{1+x^2}$ has three points of inflection and they all lie on one straight line.

$$\begin{aligned}
f'(x) =& \frac{(1+x^2) - 2x(1+x)}{(1+x^2)^2} = \frac{-x^2 - 2x + 1}{(1+x^2)^2} \\
f''(x) =& \frac{(1+x^2)^2(-2x-2) - 4x(-x^2-2x+1)(1+x^2)}{(1+x^2)^4} \\
=& \frac{(1+x^2)(-2x^3 - 2x^2 - 2x - 2 + 4x^3 + 8x^2 - 4x)}{(1+x^2)^4} \\
=& \frac{(1+x^2)(2x^3 + 6x^2 - 6x - 2)}{(1+x^2)^4} \\
f''(x) =& 0 \implies x^3 + 3x^2 - 3x - 1 = 0
\end{aligned}$$

$$x = 1 \implies (x-1)(x^2 + 4x + 1) = 0$$
$$x = \frac{-4 \pm 2\sqrt{3}}{2}$$

$$\boxed{x = 1, -2 \pm \sqrt{3}}$$

$$\begin{cases} f(1) = 1 \\ f(-2+\sqrt{3}) = \dfrac{\sqrt{3}+1}{4} \\ f(-2-\sqrt{3}) = \dfrac{1-\sqrt{3}}{4} \end{cases}$$

$$\frac{dy}{dx} = \frac{f(1) - f(-2+\sqrt{3})}{1-(-2+\sqrt{3})} = \frac{\frac{3-\sqrt{3}}{4}}{3-\sqrt{3}} = \frac{1}{4}$$

$$\frac{dy}{dx} = \frac{f(1) - f(-2-\sqrt{3})}{1-(-2-\sqrt{3})} = \frac{\frac{3+\sqrt{3}}{4}}{3+\sqrt{3}} = \frac{1}{4}$$

$$\boxed{\text{Three points of inflection are collinear}}$$

Problem 15

Suppose f is differentiable on an interval I and $f'(x) > 0$ for all numbers x in I except for a single number c. Prove that f is increasing on the entire interval I.

Given $f(x)$ is increasing on an interval I except for a single value c, as $f'(x) > 0$ on these intervals, let x_1, x_2 be arbitrary values on I s.t. $x_1 < x_2$.

For $x_1 < x_2 \leq c$, as $f(x)$ is increasing on (x_1, c), $f(x_1) < f(x_2)$.

For $c \leq x_1 < x_2$, as $f(x)$ is increasing on (c, x_2), $f(x_1) < f(x_2)$.

For $x_1 < c < x_2$, as $f(x)$ is increasing on (x_1, c) and (c, x_2), then $f(x_1) < f(c)$ and $f(c) < f(x_2)$. Thus, $f(x_1) < f(x_2)$.

Therefore, $f(x_1) < f(x_2)$ for all $x_1 < x_2$ on I.